Project — Week 1

Project Title: House Price Prediction

Assigned Date: [17/06/2026] **Submission Date:** [21/06/2026]

Problem Statement :

Real estate buyers and sellers often rely on guesswork or outdated comparisons to estimate a property's fair value. Your task is to build a regression model that predicts house prices based on property features such as size, number of rooms, location, and age — and then identify which features most strongly influence price.

Findings Summary :

What Drives Price the Most :

The two main factors determining house price are property area and number of bathrooms, according to both correlation coefficient calculation and importance ranking by the Random Forest model. The remaining three significant factors are air conditioning, number of floors, and parking lot capacity. Hot water heating and whether the home is furnished play only a marginal role in pricing.

Model Performance :

Metric	Linear Regression	Random Forest
MAE	₹970,043	₹1,014,947
RMSE	₹1,324,507	₹1,399,769
R^2 Score	0.653	0.612

Simply stated, the model accounts for approximately 65% of the variance in price difference between houses, and misses on average by about 970,000. Useful for an approximation, but still too much variation to use for determining the final price of a house without human intervention.

Surprising about data :

Random Forest, despite being more versatile, did not beat simple Linear Regression. With just over 545 records, there is simply not enough data for the more complex Random Forest to work with. This is an important lesson not all more complicated models will yield better results.

Recommendation for a Real Estate Business :

As area and number of bathrooms determine prices significantly more than other features, when marketing properties, it is important to put area and number of bathrooms as key value propositions, and not the secondary factors such as hot water heating or furnishings.